Business Process Prediction, Simulation and Optimization

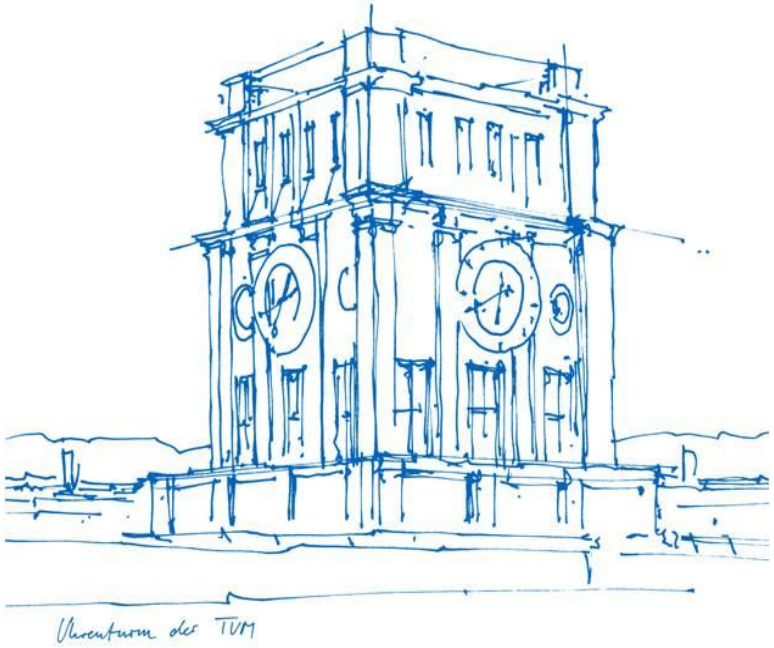
2. Process Mining I

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Garching, 22 October 2026



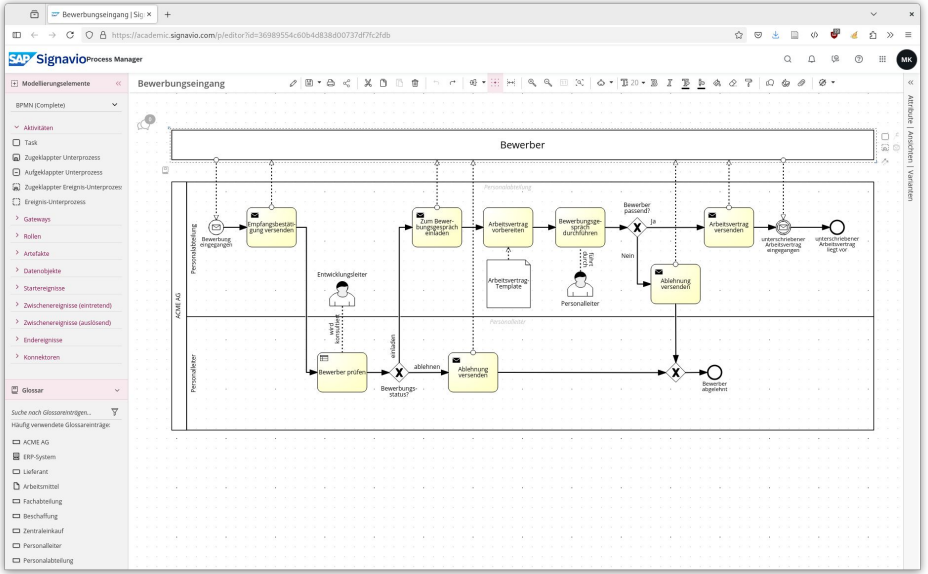
Business Process Management

A strong modeling tradition, and a free tool for students

BPM has a strong modeling history!

- Use it to model your final BPM

> SAP Signavio (free for students):
> https://academic.signavio.com/p/login



Business Process Management

A strong modeling tradition, and a backup modeler if Signavio fails

> Use **bpmn.io** if the SAP Signavio Modeler does not work
> https://demo.bpmn.io



Open or **create** a BPMN diagram.

Overview

From simulation to the BPIC 2017 process

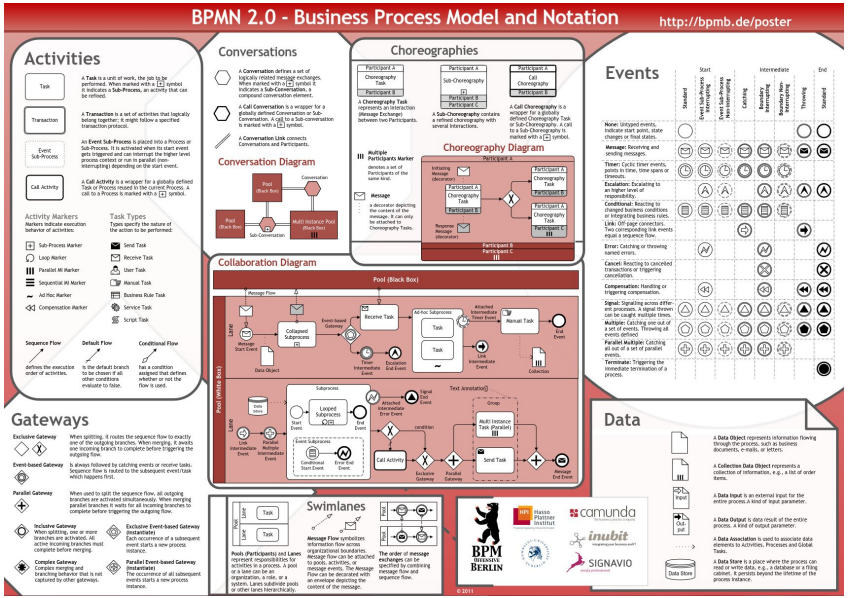
General idea: Data-driven Business Process Simulation works because we know ‘how operational business processes look like’.

Question: So how does the BPIC 2017 process look like?

Modeling Operational Processes

Business Process Model and Notation

BPMN

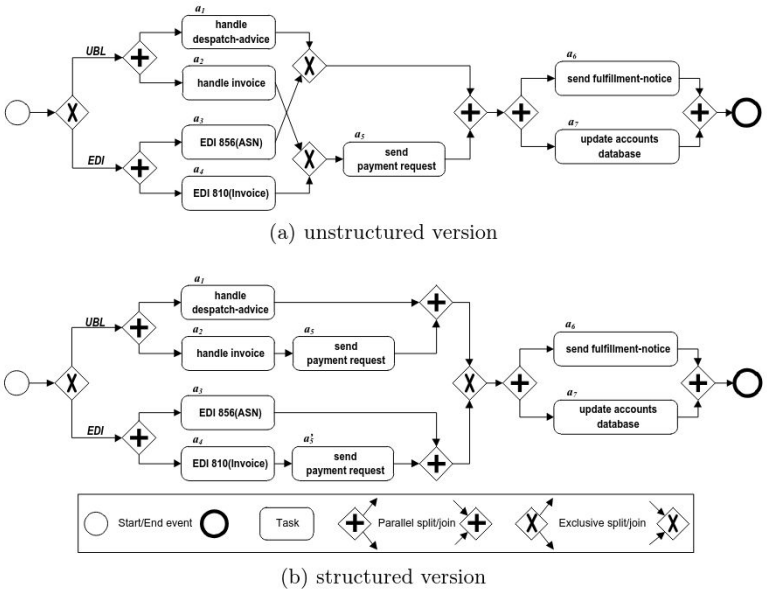


Source: https://bpmd.de/poster

Block-Structured Process Models

Definition and limitation

For every node with multiple outgoing arcs (a **split**) there is a corresponding node with multiple incoming arcs (a join), such that the subgraph between the split and the join forms a **single-entry-single-exit (SESE)** region.



(a) unstructured version

(b) structured version

> Not all unstructured process models can be transformed into equivalent structured models.

Source: [5]

From BPMN to Petri Nets

Translation

How can BPMN be translated into a Petri net?

- **Task** → transition
- **Start / end event** → source / sink place
- **Sequence flow** → place
- **XOR gateway** → place with several output (split) or input (join) arcs
- **AND gateway** → transition with several output (split) or input (join) arcs

Is the translation done without information loss?

- **Not entirely.** Control flow is preserved for tasks, events, XOR, and AND gateways
- **Lost:** layout, e.g., pools or swimlanes

Petri Net Based Modeling

Four reasons to model processes with Petri nets

Why Petri nets?



Graphical representation



Formal semantics



Expressiveness



Formal analysis

Petri Nets

Places, transitions, flow relation, and initial marking

$$N = (P, T, F, M_0)$$

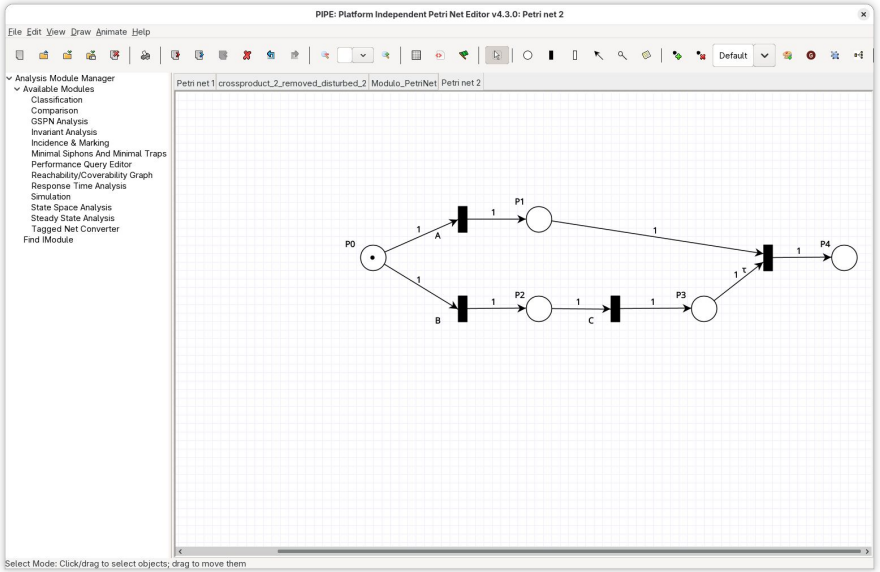
P : Places

T : Transitions

F : $F \subseteq (P \times T) \cup (T \times P)$

M_0 : Initial Marking

$M : P \mapsto \mathbb{N}$



Petri Net Based Modeling

Extensions add expressiveness, but processes rarely need it

Extensions

- High-level Petri nets are often Turing complete (Type-0 languages)
- Many extensions of classical Petri nets exist
  - e.g., Colored Petri nets

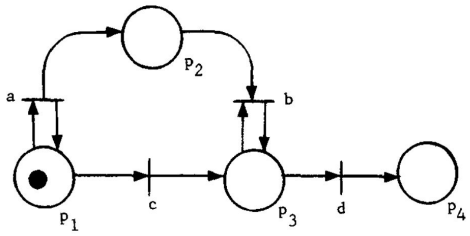
Modeling Tradeoff

- Modeling: Tradeoff between expressiveness and decidability
- Petri nets are sometimes unnecessary expressive for processes
- Workflow nets are often sufficient

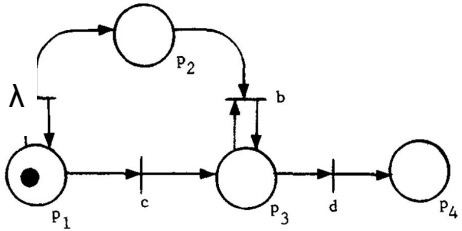
Expressiveness of Petri Net Languages

Labels and final states determine expressive power

- A Petri net and a marking of that Petri net can be used to define a **Petri net language** (all possible execution sequences)
- The expressive power of these languages depend on (see [1]):
  - Allowed transition labels (arbitrary, λ-free, free)
  - The definition of the set of final states (L-type, G-type, T-type, P-type)



$\{a^n c b^n d | n \geq 0\}$



$\{cb^\star d\}$

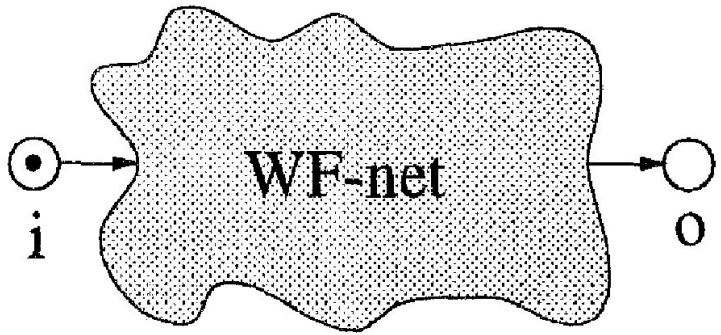
Source: [1]

Workflow Nets

A subclass of Petri nets with one source, one sink, and connected nodes

Workflow nets offer a subclass of Petri nets

- A Petri net is a workflow net iff:
  - There is a single source place i
  - There is a single sink place o
  - Every node is on a path from i to o



Source: [2]

Workflow Nets

Soundness guarantees that every case terminated properly

Desirable property of a workflow net: **soundness**

> “For any case, the procedure will terminate eventually, and at the moment the procedure terminates there is a token in place o and all the other places are empty.” [2]

A workflow net is sound if:

Option to complete	For every state M reachable from i, there exists a firing sequence leading from M to o.
Proper completion	When o contains a token, all other places are empty
Dead transitions	There are no dead transitions

Workflow Nets

Definitions of soundness and boundedness, and how they relate

Sound: every case ends properly

- **Option to complete:** o can be reached from every reachable state
- **Proper completion:** a token in o means all other places are empty
- **No dead transitions:** every transition can fire

Bounded: finite tokens per place

- **Formally:** for some fixed k, $M(p) \leq k$ for every place p and every marking M reachable from i
- **Consequence:** finitely many reachable states

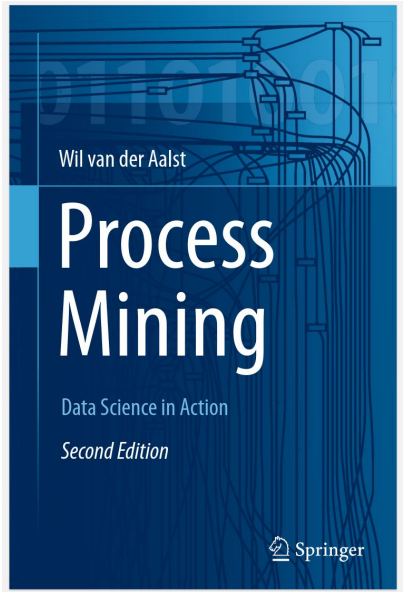
> If a workflow net is sound, then it is also bounded (for a proof see [2])

Process Mining

Recommendation

A textbook introduction to process mining

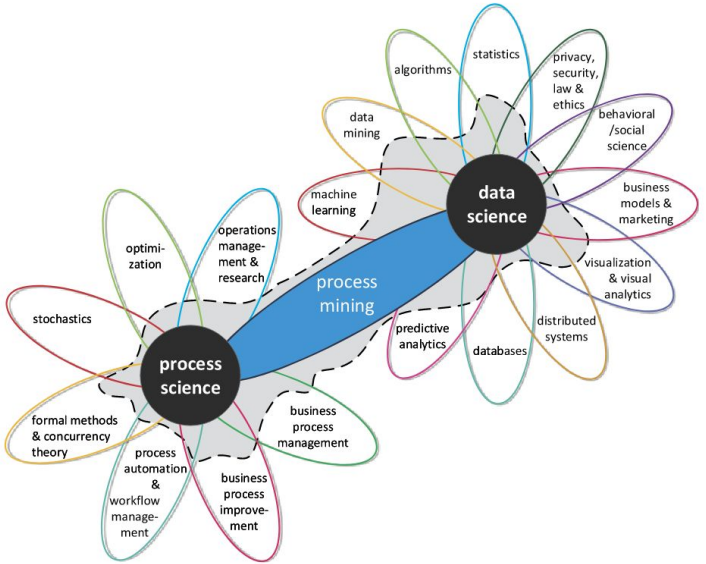
- Well written book [4]
- Introduces the basic concepts in process mining trough easy-to-follow examples



What Is Process Mining?

Definition and place in data science

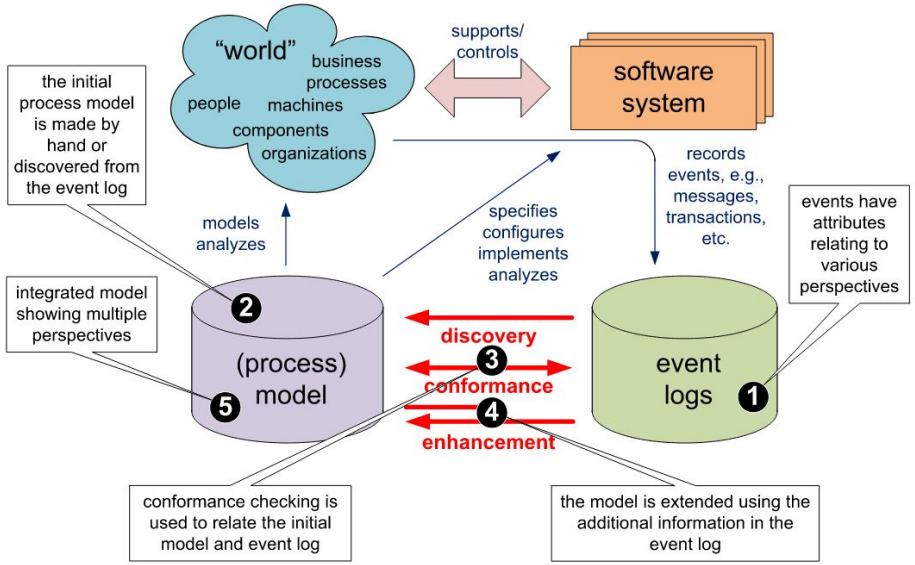
> "The goal of process mining is to turn event data into insights and actions. Process mining is an integral part of data science, fueled by the availability of data and the desire to improve processes." [4, p. 2]



Source: [1]

Main Types of Process Mining

Discovery, conformance checking, and enhancement



Source: [4]

Three Main Types

What each type does with an event log

Discovery	Produces a model based on an event log without using any a-priori information
Conformance	An existing process model is compared with an event log of the same process
Enhancement	Improve an existing process model using information about the actual process recorded in an event log

Event Logs

Event Log

Structure of an event log

Case id	Event id	Properties				
		Timestamp	Activity	Resource	Cost	...
1	35654423	30-12-2010:11.02	register request	Pete	50	...
	35654424	31-12-2010:10.06	examine thoroughly	Sue	400	...
	35654425	05-01-2011:15.12	check ticket	Mike	100	...
	35654426	06-01-2011:11.18	decide	Sara	200	...
	35654427	07-01-2011:14.24	reject request	Pete	200	...
2	35654483	30-12-2010:11.32	register request	Mike	50	...
	35654485	30-12-2010:12.12	check ticket	Mike	100	...
	35654487	30-12-2010:14.16	examine casually	Pete	400	...
	35654488	05-01-2011:11.22	decide	Sara	200	...
	35654489	08-01-2011:12.05	pay compensation	Ellen	200	...
3	35654521	30-12-2010:14.32	register request	Pete	50	...
	35654522	30-12-2010:15.06	examine casually	Mike	400	...
	35654524	30-12-2010:16.34	check ticket	Ellen	100	...
	35654525	06-01-2011:09.18	decide	Sara	200	...
	35654526	06-01-2011:12.18	reinitiate request	Sara	200	...
	35654527	06-01-2011:13.06	examine thoroughly	Sean	400	...
	35654530	08-01-2011:11.43	check ticket	Pete	100	...
	35654531	09-01-2011:09.55	decide	Sara	200	...
	35654533	15-01-2011:10.45	pay compensation	Ellen	200	...
4	35654641	06-01-2011:15.02	register request	Pete	50	...
	35654643	07-01-2011:12.06	check ticket	Mike	100	...

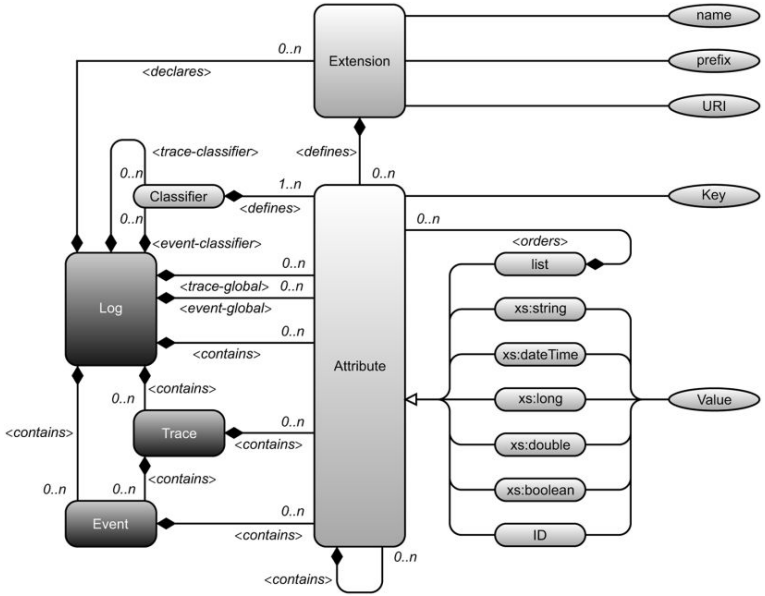
Event Log

XES is the IEEE standard format for event logs

Format	1849-2023 - IEEE Standard for eXtensible Event Stream (XES) for Achieving Interoperability in Event Logs and Event Streams https://doi.org/10.1109/IEEESTD.2023.10267858
File types	xml, yaml, json, csv, …

XES Event Log

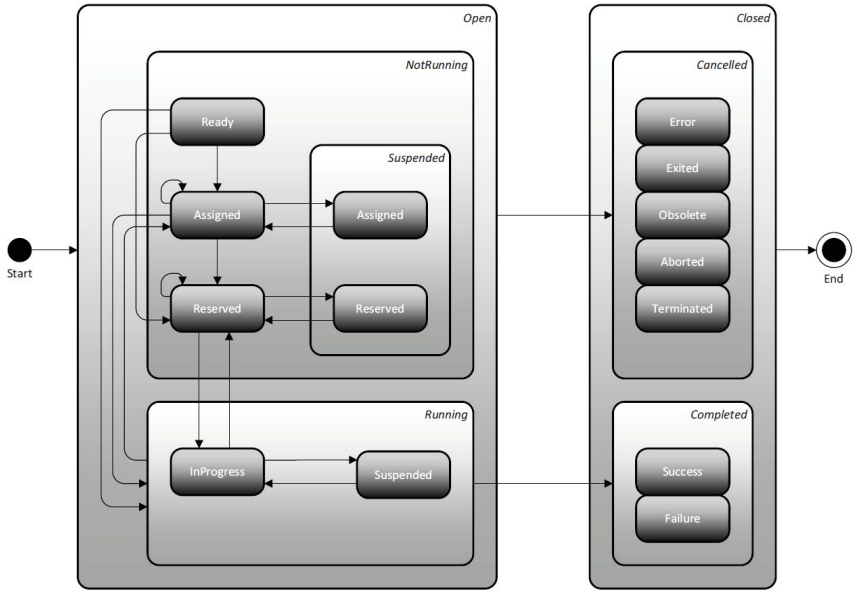
Structure of the XES standard format



Source: [5]

Lifecycle Extension

XES standard extension for recording activity state transitions



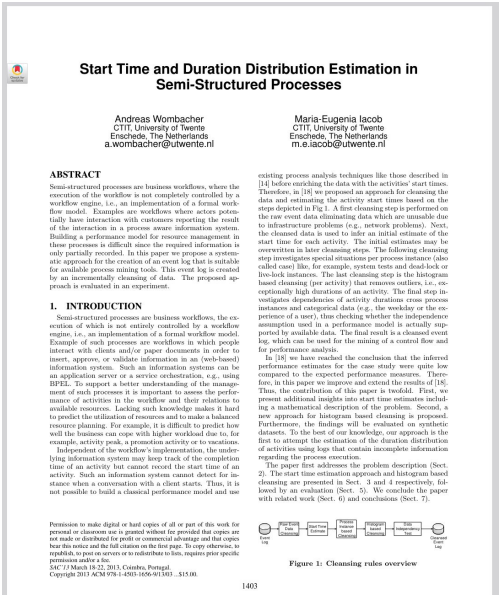
Source: https://www.tf-pm.org/resources/xes-standard/about-xes/standard-extensions/lifecycle/bpaf

Why Is the Lifecycle Extension Important?

Start and end events separate processing time from waiting time

- When only the start or end time of an activity is tracked, we can not distinguish the processing time of an activity with a preceding waiting time
  - although workarounds exist (see [6])

> ⚠Danger⚠: Even with the lifecycle information the durations might not represent actual processing of an activity (e.g. because the activity was started in the afternoon and finished in the next morning)



Start Time and Duration Distribution Estimation in Semi-Structured Processes

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ABSTRACT

Semi-structured processes are business workflows, where the execution of the workflow is not completely controlled by a workflow engine, i.e., an implementation of a formal workflow model. Examples are workflows where actors potentially have interaction with customers reporting the result of the interaction in a process aware information system. Building a performance model for resource management in these processes is difficult since the required information is only partially recorded. In this paper we propose a systematic approach for the creation of an event log that is suitable for available process mining tools. This event log is created by an incrementally cleansing of data. The proposed approach is evaluated in an experiment.

1. INTRODUCTION

Semi-structured processes are business workflows, the execution of which is not entirely controlled by a workflow engine, i.e., an implementation of a formal workflow model. Example of such processes are workflows in which people interact with clients and/or paper documents in order to insert, approve, or validate information in an (web-based) information system. Such an information systems can be an application server or a service orchestration, e.g., using BPEL. To support a better understanding of the management of such processes it is important to assess the performance of activities in the workflow and their relations to available resources. Lacking such knowledge makes it hard to predict the utilization of resources and to make a balanced resource planning. For example, it is difficult to predict how well the business can cope with higher workload due to, for example, activity peak, a promotion activity or to vacations.

Independent of the workflow's implementation, the underlying information system may keep track of the completion time of an activity but cannot record the start time of an activity. Such an information system cannot detect for instance when a conversation with a client starts. Thus, it is not possible to build a classical performance model and use existing process analysis techniques like those described in [14] before enriching the data with the activities' start times. Therefore, in [18] we proposed an approach for cleansing the data and estimating the activity start times based on the steps depicted in Fig 1. A first cleansing step is performed on the raw event data eliminating data which are unusable due to infrastructure problems (e.g., network problems). Next, the cleansed data is used to infer an initial estimate of the start time for each activity. The initial estimates may be overwritten in later cleansing steps. The following cleansing step investigates special situations per process instance (also called case) like, for example, system tests and dead-lock or live-lock instances. The last cleansing step is the histogram based cleansing (per activity) that removes outliers, i.e., exceptionally high durations of an activity. The final step investigates dependencies of activity durations cross process instances and categorical data (e.g., the weekday or the experience of a user), thus checking whether the independence assumption used in a performance model is actually supported by available data. The final result is a cleansed event log, which can be used for the mining of a control flow and for performance analysis.

In [18] we have reached the conclusion that the inferred performance estimates for the case study were quite low compared to the expected performance measures. Therefore, in this paper we improve and extend the results of [18]. Thus, the contribution of this paper is twofold. First, we present additional insights into start time estimates including a mathematical description of the problem. Second, a new approach for histogram based cleansing is proposed. Furthermore, the findings will be evaluated on synthetic datasets. To the best of our knowledge, our approach is the first to attempt the estimation of the duration distribution of activities using logs that contain incomplete information regarding the process execution.

The paper first addresses the problem description (Sect. 2). The start time estimation approach and histogram based cleansing are presented in Sect. 3 and 4 respectively, followed by an evaluation (Sect. 5). We conclude the paper with related work (Sect. 6) and conclusions (Sect. 7).

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SAC'13 March 18-22, 2013, Coimbra, Portugal.
Copyright 2013 ACM 978-1-4503-1656-9/13/03 ...$15.00.

Figure 1: Cleansing rules overview

Process Discovery

Process Mining

Process Discovery

- **discovery:** produces a model based on an event log without using any a-priori information.
- **conformance:** An existing process model is compared with an event log of the same process.
- **enhancement:** Improve an existing process model using information about the actual process recorded in an event log.

Process Discovery Is Challenging

Logs rarely contain every possible behavior of the process

- Data sparsity
- Parallels

> "consider a process consisting of 10 activities that can be executed in parallel and a corresponding log that contains information about 10,000 cases. The total number of possible interleavings in the model with 10 concurrent activities is 10! = 3,628,800. Hence, it is impossible that each interleaving is present in the log as there are fewer cases (10,000) than potential traces (3,628,800). Even if there are 3,628,800 cases in the log, it is extremely unlikely that all possible variations are present." [4]

Process Discovery Algorithms

Alpha Miner and Heuristics Miner can return unsound workflow nets

Alpha Miner

- Creates a footprint matrix
- Can be converted to a workflow net, but produces unsound nets on many real logs

Heuristics Miner

- DFG with dependency scores
- Can also be converted to a workflow net, but also not guaranteed to be sound

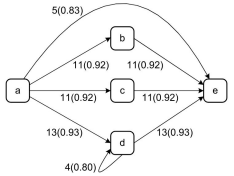
Table 6.1 Footprint of L_1: $a \#_{L_1} a$, $a \rightarrow_{L_1} b$, $a \rightarrow_{L_1} c$, etc.

	a	b	c	d	e
a	$\#_{L_1}$	$\rightarrow_{L_1}$	$\rightarrow_{L_1}$	$\#_{L_1}$	$\rightarrow_{L_1}$
b	$\leftarrow_{L_1}$	$\#_{L_1}$	$\|\|_{L_1}$	$\rightarrow_{L_1}$	$\#_{L_1}$
c	$\leftarrow_{L_1}$	$\|\|_{L_1}$	$\#_{L_1}$	$\rightarrow_{L_1}$	$\#_{L_1}$
d	$\#_{L_1}$	$\leftarrow_{L_1}$	$\leftarrow_{L_1}$	$\#_{L_1}$	$\leftarrow_{L_1}$
e	$\leftarrow_{L_1}$	$\#_{L_1}$	$\#_{L_1}$	$\rightarrow_{L_1}$	$\#_{L_1}$

Table 7.2 Dependency measures between the five activities based on event log L

$\|\Rightarrow_L\|$	a	b	c	d	e
a	$\frac{0}{0+1} = 0$	$\frac{11-0}{11+0+1} = 0.92$	$\frac{11-0}{11+0+1} = 0.92$	$\frac{13-0}{13+0+1} = 0.93$	$\frac{5-0}{5+0+1} = 0.83$
b	$\frac{0-11}{0+11+1} = -0.92$	$\frac{0}{0+1} = 0$	$\frac{10-10}{10+10+1} = 0$	$\frac{0-0}{0+0+1} = 0$	$\frac{11-0}{11+0+1} = 0.92$
c	$\frac{0-11}{0+11+1} = -0.92$	$\frac{10-10}{10+10+1} = 0$	$\frac{0}{0+1} = 0$	$\frac{0-0}{0+0+1} = 0$	$\frac{11-0}{11+0+1} = 0.92$
d	$\frac{0-13}{0+13+1} = -0.93$	$\frac{0-0}{0+0+1} = 0$	$\frac{0-0}{0+0+1} = 0$	$\frac{4}{4+1} = 0.80$	$\frac{13-0}{13+0+1} = 0.93$
e	$\frac{0-5}{0+5+1} = -0.83$	$\frac{0-11}{0+11+1} = -0.92$	$\frac{0-11}{0+11+1} = -0.92$	$\frac{0-13}{0+13+1} = -0.93$	$\frac{0}{0+1} = 0$

Fig. 7.4 Dependency graph using a threshold of 2 for $|>_L|$ and 0.7 for $|\Rightarrow_L|$: *each arc* shows the $|>_L|$ value and the $|\Rightarrow_L|$ value between brackets. For example, $|a >_L d| = 13$ and $|a \Rightarrow_L d| = 0.93$



Source: [4]

Process Discovery Algorithms

The Inductive Miner always returns a sound, block-structured model

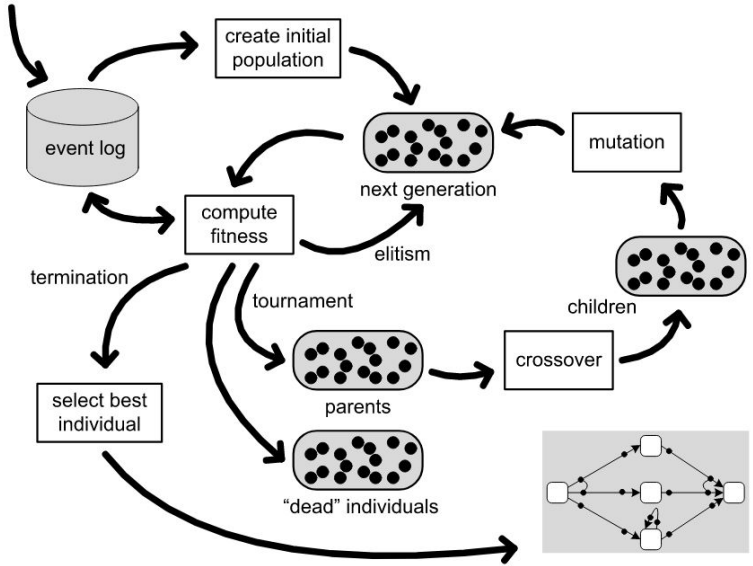
Inductive Miner

- Recursively splits the event log along one of four cut types, then repeats on each sub-log
- Returns a process tree, which converts to a sound, block-structured workflow net
- Has (with temperature 0), by design a fitness of 1.

Exclusive choice	Each case follows exactly one of the branches
Sequence	The branches run one after the other
Parallel	The branches run in any order or interleaved
Loop	A body part repeats, with a redo part in between

Genetic Miner

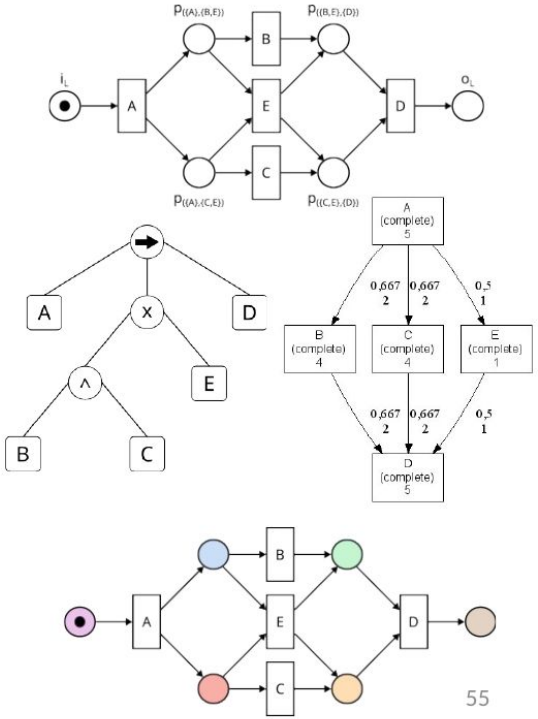
Discovers process models by evolutionary search



Source: [4]

Ergebnisse/Herausforderungen

- Ergebnisse für L = {<A,B,C,D>,<A,C,B,D>,<A,B,C,D>,<A,C,B,D>,<A,E,D>}
  - α-Miner Eigenschaften:
    - Garantiert nicht, dass Modell alle Spuren im Log erzeugen kann
    - Probleme mit “short-loops” (Schleifen mit Länge 1 oder 2)
  - Heuristic Miner Eigenschaften:
    - Garantiert nicht, dass Modell alle Spuren im Log erzeugen kann
    - Fokus darauf mit noisy und unvollständigen Daten umzugehen
  - Inductive Miner Eigenschaften:
    - Garantiert, dass alle Spuren im Log erzeugt werden können
    - Mehrere Prozessbäume können log-äquivalent sein - z.B. ×(A,B,C) und ×(C,×(B,A)) – aber nur ein spezifischer kann vom Inductive Miner discovered werden
  - Genetic Miner Eigenschaften:
    - Garantiert nicht, dass Modell alle Spuren im Log erzeugen kann
    - Für größere Modelle und Logs nicht sehr effizient
  - Region-Based Miner Eigenschaften:
    - Kann je nach Zustandsrepräsentation zu einem Modell führen, das over- oder underfitted
- Generelle Herausforderungen für alle Miner:
  - Wie werden Daten gesammelt (und evtl. gefiltert) um den initialen Log zu erzeugen??
  - Reagieren sensitiv auf noisy und unvollständige Daten (außer Heustic Miner)
  - Gleiche Labels bei verschiedenen Aktivitäten führen zu Problemen



Hands-On

Loan Application Process

The event log used throughout the hands-on part

Dataset from the BPI Challenge 2017 (BPIC 2017)

Download the event log:

https://doi.org/10.4121/uuid:5f3067df-f10b-45da-b98b-86ae4c7a310b



Photo: https://commons.wikimedia.org/wiki/File:2020_ING_Bijlmerdreef_106_(1).jpg

Setup Your Tools

A Python environment with Pm4Py, pandas, and scikit-learn

- Set up Python / PIP / Pipenv / Jupyter notebooks
- Use Pm4Py to load the xes.xml event log into a pandas dataframe
  - https://pypi.org/project/pm4py/
- Use Pandas' select operations and look into the event log, e.g. individual cases, resources, etc.
- Use scikit-learn for basic data science and machine learning







Pm4Py

Useful tools, but plan to patch the package locally

- Pm4Py has implemented a lot of helpful tools which we will be useful throughout this course
- But it’s code is terrible
- Therefore, it might be necessary that you make some fixes to the packages code



Assignment I

Process identification, discovery, and insights

> Individual assignment counts 25% on total grade.

What to do?

Deliverable	Prepare a report PDF (use the template, 5 pages) and set up Github or Gitlab repository with your code (public, link in file).
Submission	19 November 2026, 23:59 on Moodle

Tasks: Understand What Is Going On!

Tasks 1 and 2: read the literature, then explore the data

1) Read Available Descriptions and Reports

- Whatever you find!
- e.g.: https://ais.win.tue.nl/bpi/2017/challenge.html
- Present an overview in the report - state of the art and limitations.

2) Get Into the Data Set

- Load it with pm4py.
- How does a case typically look like?
- How do long cases look like? How do short ones look like?

Tasks: Understand What Is Going On!

Tasks 3 and 4: compute metrics and visualize them

3) Calculate Basic Metrics About the Process Using Python / Pm4Py

- Number of cases
- Number of process variants
- Number of events / activities / resources
- Distributions of attributes
- Average cycle times of cases / activities (+ std. dev.)
- …

4) Provide meaningful visualizations.

Tasks: Understand What Is Going On!

Tasks 5 and 6: mine the process and analyze it further

5) Process Mining

- Apply process discovery, conformance checking (and decision mining, only mandatory for Masters in Advanced course).
- Refine the process model and submit your final BPMN with quality metrics.
- Reason and explain your design decisions.

6) Perform an advanced analysis.

Literature

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Literature

References, continued

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